

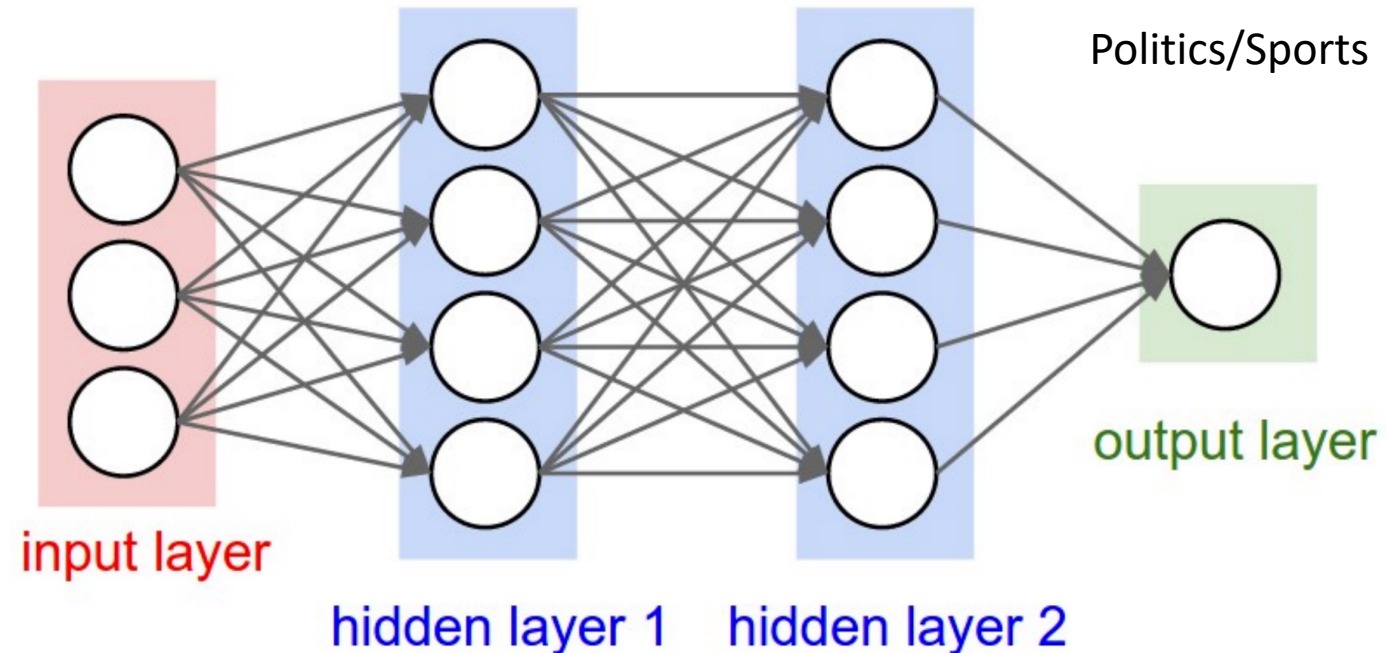
# Prediction-based Models

- Idea: instead of directly representing the distribution of a word, we can represent words as a vector from which the distribution of a word can be “decoded”
- **Task:** learn a network to predict a neighboring word from a given word
  - Sometimes called “self-supervision”
- An influential suite of methods for prediction-based embeddings is *word2vec* (Mikolov et al., 2013)
- We will review the basic implementation of the skip-gram model

# Word Embeddings as Features

- Word embeddings are often used as features to supervised learning tasks
- For instance, in text classification:

Instead of bag-of-words, one might input the sum of the embeddings of the words in the document



# Word Embeddings as Features

- In this case, we don't really care what the dimensions represent
  - They are just useful feature representations, where the “semantics” of these features are unknown
- In fact, word embeddings are often used as initialization to the first layer of a neural network, and are later updated during training
  - This is sometimes called “fine-tuning”
  - Later in the course...
- Applications cover all aspects of NLP
  - We will see an example next lesson, and more towards the end of the course
- The central limitation of skip-gram is its non-contextualization. We will revisit this issue later in the course